Languages, automata and computation II
Tutorial 10 – Ideals, varieties, and polynomial automata

Winter semester 2024/2025

In this tutorial we explore ideals, varieties, and polynomial automata. Recall that an *ideal* of a ring R is a subset $I \subseteq R$ which is 1) closed under sum $I + I \subseteq I$, and 2) closed under under product with elements from the ring $R \cdot I \subseteq I$. For a set of vectors $A \subseteq \overline{\mathbb{Q}}^k$, let $I(A) \subseteq \overline{\mathbb{Q}}[x_1, \ldots, x_k]$ be the set of polynomials vanishing on A. (This is an ideal, justifying the notation). For a set of polynomials $P \subseteq \overline{\mathbb{Q}}[x_1, \ldots, x_k]$, let $V(P) \subseteq \overline{\mathbb{Q}}^k$ be the set of vectors where all polynomials in P vanish simultaneously. The *Zariski closure* of a set of vectors A is defined as

$$\overline{A} := V(I(A)).$$

Exercise 1.
1. Show that $A \subseteq \overline{A}$, for every $A \subseteq \overline{\mathbb{Q}}^k$.
2. Find a set of vectors $A \subseteq \overline{\mathbb{Q}}^k$ where the inclusion in the previous point is strict. Can such a set A be finite?

Exercise 2 (zero polynomial vs. zero polynomial function)**.** Show that $p : \overline{\mathbb{Q}}[x_1, \ldots, x_k]$ is the zero polynomial iff as a function $\overline{\mathbb{Q}}^k \to \overline{\mathbb{Q}}$ it is constantly zero. Is this true if we replace $\overline{\mathbb{Q}}$ by $\mathbb{F}_2$ (the field consisting just of the elements $\{0, 1\}$)?

Exercise 3. Show that I is an ideal of $R[x]$ iff I is a vector subspace of $R[x]$ over R (i.e., $I + I \subseteq I$ and $aI \subseteq I$ for every $a \in R$) s.t. $xI \subseteq I$.

Exercise 4. We have seen that $I(A)$ is an ideal of $\overline{\mathbb{Q}}[x_1, \ldots, x_k]$ for every $A \subseteq \overline{\mathbb{Q}}^d$. Is every ideal of this ring of this form?

Principal ideal rings

Recall that a ring R is a *principal ideal ring* if every ideal of R is generated by one element.

Exercise 5. Are the following principal ideal rings?

1. The field of rational numbers $\mathbb{Q}$.
2. The ring of integers $\mathbb{Z}$.

3. The ring of univariate polynomials over the rationals $\mathbb{Q}[x]$.

4. The ring of univariate polynomials over the integers $\mathbb{Z}[x]$.

5. The ring of bivariate polynomials over the rationals $\mathbb{Q}[x, y]$.

6. The quotient ring $\mathbb{Q}[x, y]/\langle x - y \rangle$.

Exercise 6. If R is a principal ideal ring, does the same hold for $R[x]$?

Noetherian rings

A ring R is *Noetherian* if every ideal $I \subseteq R$ is finitely generated. In the following problem we explore ways to construct Noetherian rings.

Exercise 7. 1. Fields are Noetherian.

2. Finite rings are Noetherian.

3. Principal ideal rings are Noetherian.

4. If R is Noetherian and $I \subseteq R$ is an ideal, then R/I is Noetherian.

5. If R is Noetherian, then $R[x]$ is Noetherian. Does the converse hold?

6. If R is Noetherian, then $R[\![x]\!]$ is Noetherian.

Exercise 8. Are the following rings Noetherian?

1. Ring of polynomials with countably many variables: $\mathbb{Q}[x_1, x_2, \dots]$.

2. Ring of power series: $\mathbb{Q}[\![x]\!]$.

3. Ring of rational power series: $\mathbb{Q}[\![x]\!] \cap \mathbb{Q}(x)$.

4. Noncommutative ring of power series in noncommuting variables: $R := \Sigma^* \to \mathbb{Q}$, $|\Sigma| \geq 2$, with sum and convolution (Cauchy) product.

Exercise 9. For every of the following sets A check that it is a ring. Is it Noetherian?

1. $A = x \cdot \mathbb{Q}[x] \subseteq \mathbb{Q}[x]$.

2. $A = \mathbb{Z} + x \cdot \mathbb{Q}[x] \subseteq \mathbb{Q}[x]$.

Polynomial automata

Recall that a *polynomial automaton* is a tuple

$$A = (d, \Sigma, Q, q_I, p, F)$$

where $d \in \mathbb{N}$ is the *dimension*, Σ is a finite alphabet, $Q = \overline{\mathbb{Q}}^d$ is the set of *states*, $q_I \in Q$ is the *initial state*, $p : \Sigma \to \mathbb{Q}[d]^d$ is a collection of tuples of polynomials inducing a polynomial action on states

$$q \in Q \mapsto q \cdot a \in Q, \quad \text{for every } a \in \Sigma,$$

where $q \cdot a := p^a(q) = (p_1^a(q), \ldots, p_d^a(q))$, and $F : Q \to \overline{\mathbb{Q}}$ is the polynomial *output function*. The action of Σ is extended to words $w \in \Sigma^*$ homomorphically: $q \cdot \varepsilon := q$ and $q \cdot (a \cdot w) := (q \cdot a) \cdot w$. The *semantics* of state $q \in Q$ is the mapping $[\![q]\!] : \Sigma^* \to \overline{\mathbb{Q}}$ defined as

$$[\![q]\!]_w = F(q \cdot w), \quad \text{for every } w \in \Sigma^*.$$

The semantics of the automaton A is $[\![A]\!] = [\![q_I]\!]$. The automaton is *zero* if $[\![A]\!] = 0$.

The set $\Sigma^* \to \mathbb{Q}$ has the structure of a commutative ring w.r.t. addition and Hadamard product (pointwise product). This gives us an alternative presentation of the semantics of polynomial automata.

Exercise 10. A polynomial automaton A over a unary alphabet $\{a\}$ computes a function $f : \mathbb{N} \to \mathbb{N}$ if $[\![A]\!]_{a^n} = f(n)$ for $n \in \mathbb{N}$. Give examples of polynomial automata over a unary alphabet that compute the following functions:

- $f(n) = n!$
- $f(n) = 2^{2^n}$
- any rational function (computable by a unary weighted automaton).

Exercise 11. Show that a function $f : \Sigma^* \to \mathbb{Q}$ is recognisable by a polynomial automaton iff its reversal f^R belongs to a finitely generated subring of $\Sigma^* \to \mathbb{Q}$ closed under left quotients $u^{-1}(\cdot)$.

Exercise 12. Consider the following computational model B. States are tuples of polynomials $S = \mathbb{Q}[d]^d$, with $p_I := (x_1, \ldots, x_d)$ being the initial state and $F : S \to \mathbb{Q}$ a polynomial output function. The update function is described by a tuple of polynomials $p^a = (p_1^a, \ldots, p_d^a) \in \mathbb{Q}[d]^d$, one for each $a \in \Sigma$, by polynomial substitution as follows:

$$p \mapsto p \cdot a := p(p^a) \in S, \quad \text{for every } a \in \Sigma.$$

In other words, in the current state p we replace x_1 by p_1^a, $\ldots$, x_d by p_d^a. This is extended homomorphically to $\Sigma^* \to S$. For instance $p_I \cdot ab = p^a(p^b)$ and $p_I \cdot abc = p^a(p^b)(p^c) = p^a(p^b(p^c))$. The output on reading w is $[\![B]\!]_w = F(p_I \cdot w)$. Decide zeroness for B.

A *variety* is a subset $V \subseteq \overline{\mathbb{Q}}^d$ which is the set of common zeros of a set of polynomials: $V = V(P)$ for some $P \subseteq \mathbb{Q}[x_1, \ldots, x_d]$.

Exercise 13. For $d = 2$ find a non-trivial infinite variety.

Exercise 14. Let $V \subseteq \overline{\mathbb{Q}}^d$ be a variety.

1. Let $g : \overline{\mathbb{Q}}^e \to \overline{\mathbb{Q}}^d$ be a polynomial map. Is $g^{-1}(V) \subseteq \overline{\mathbb{Q}}^e$ a variety?
2. Let $g : \overline{\mathbb{Q}}^d \to \overline{\mathbb{Q}}^e$ be a polynomial map. Is $g(V) \subseteq \overline{\mathbb{Q}}^e$ a variety?

Exercise 15. Show that all real varieties $V \subseteq \mathbb{R}^d$ are generated by *a single* polynomial. Is this true for varieties of $\mathbb{C}^d$?

Exercise 16. Let $U, V \subseteq \overline{\mathbb{Q}}^d$ be a varieties. Are the following varieties?

1. $U \cap V$. What about possibly infinite intersections?
2. $U \cup V$. What about possibly infinite intersections?
3. $U \setminus V$.

Exercise 17. For each direction of the statements below, prove it if it holds, or find a counter-example otherwise.

1. For sets $A, B \subseteq \overline{\mathbb{Q}}^d$: $A \subseteq B$ iff $I(B) \subseteq I(A)$.
2. For sets of polynomials $P, Q \subseteq \overline{\mathbb{Q}}[x_1, \ldots, x_d]$: $P \subseteq Q$ iff $V(Q) \subseteq V(P)$.
3. For varieties $U, V \subseteq \overline{\mathbb{Q}}^d$: $U \subseteq V$ iff $I(V) \subseteq I(U)$.

In the next problem we explore an algorithm to decide zeroness of polynomial automata.

Exercise 18. Consider the set of states $V_n \subseteq Q$ which give zero after reading words of length $\leq n$:

$$V_n = \left\{ q \in Q \mid \forall w \in \Sigma^{\leq n} : q \cdot w = 0 \right\}.$$

1. Show that the automaton is zero iff $q_I \in \bigcap_n V_n$.
2. Show that $V_0 \supseteq V_1 \supseteq \cdots \supseteq \bigcap_n V_n$ is a nonincreasing chain of varieties. Conclude that the chain stabilises at some finite level: There is $N \in \mathbb{N}$ s.t. $V_N = V_{N+1} = \cdots = \bigcap_n V_n$.
3. Show that for every $n \in \mathbb{N}$, $V_n = V_{n+1}$ implies $n = N$.
4. Let P_n be a finite set of polynomials s.t. $V_n = V(P_n)$. Show that we can compute a finite set of polynomials P_{n+1} s.t. $V_{n+1} = V(P_{n+1})$.
5. Show how to decide $V(P) = V(Q)$ for two finite set of polynomials $P, Q \subseteq \mathbb{Q}[d]$.
6. Conclude with an algorithm for zeroness.

Exercise 19. Provide a coRP algorithm (randomised polynomial time) for the following problem: Given a polynomial automaton A and an input word $w \in \Sigma^*$, decide whether $[\![A]\!]_w = 0$.

Exercise 20. Give an algorithm for the following problem: In input we are given a polynomial automaton A and a finite automaton B recognising a regular language $L \subseteq \Sigma^*$. In output we answer whether for every $w \in L$ we have $[\![A]\!]_w = 0$.